

Circuit Simulation Project

<https://esim.fossee.in/circuit-simulation-project>



The Circuit Simulation Project is an initiative of FOSSEE, IIT Bombay that promotes the use of eSim for designing and simulating electrical and electronic circuits for learning and application purposes. The objective is to create and document original, application-based, or proof-of-concept circuit designs using eSim to build an open-source resource database.

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Title of the circuit: 5V Regulated DC Power Supply Circuit using L7805 Linear Voltage Regulator

Theory/Description: This circuit converts an unregulated 9V-12V DC input into a stable, low-ripple 5V DC output using the L7805 three-terminal linear voltage regulator. An unregulated supply is first applied at input connector J1. Capacitor C2 (1000 μ F electrolytic) filters low-frequency ripple on the incoming supply, while C1 (0.1 μ F ceramic) removes high-frequency noise before the signal reaches the IN pin of U1 (L7805). Internally, the L7805 uses a bandgap voltage reference and a series-pass transistor in a feedback loop to hold its OUT pin at a fixed 5V regardless of reasonable variations in input voltage or load current, dissipating the excess input-output voltage difference as heat. On the output side, capacitor C3 (10 μ F) improves transient response and further suppresses ripple on the regulated rail. A series resistor R1 (330 Ω) limits current through LED D1, which lights up whenever the 5V rail is active, giving a simple visual confirmation that the supply is working. The regulated 5V output is made available at connector J2 for powering microcontroller boards, sensor modules, and other low-voltage digital circuits.

Reason to simulate with eSim: eSim provides a complete, open-source EDA workflow (schematic capture, footprint assignment, netlist generation, PCB layout, and 3D visualization) that makes it possible to design, verify, and physically realize this power supply without relying on proprietary tools. It also allows the DC behaviour of the regulator circuit to be cross-checked using NgSpice before committing to a physical layout, and gives hands-on experience with dual-layer PCB routing, footprint selection, and Design Rule Checking, all of which are core skills for practical power-electronics design.

Expected Outcome/outputs : When powered with a 9V-12V DC input at J1, the circuit is expected to deliver a steady 5V DC output at J2 with minimal ripple, and the indicator LED D1 should light up as soon as input power is applied. Performance can be validated by measuring the output voltage across J2 with a multimeter across the full input range (9V-12V) and confirming it stays at 5V $\pm$ the regulator's tolerance, and by checking that the output remains stable as load current is varied up to the regulator's rated capacity.

Circuit Diagram(s) : See Figure 1 below -schematic captured in eSim's Schematic Editor (eSchema), showing J1, C2, C1, U1 (L7805), C3, R1, D1, and J2 with their interconnections.

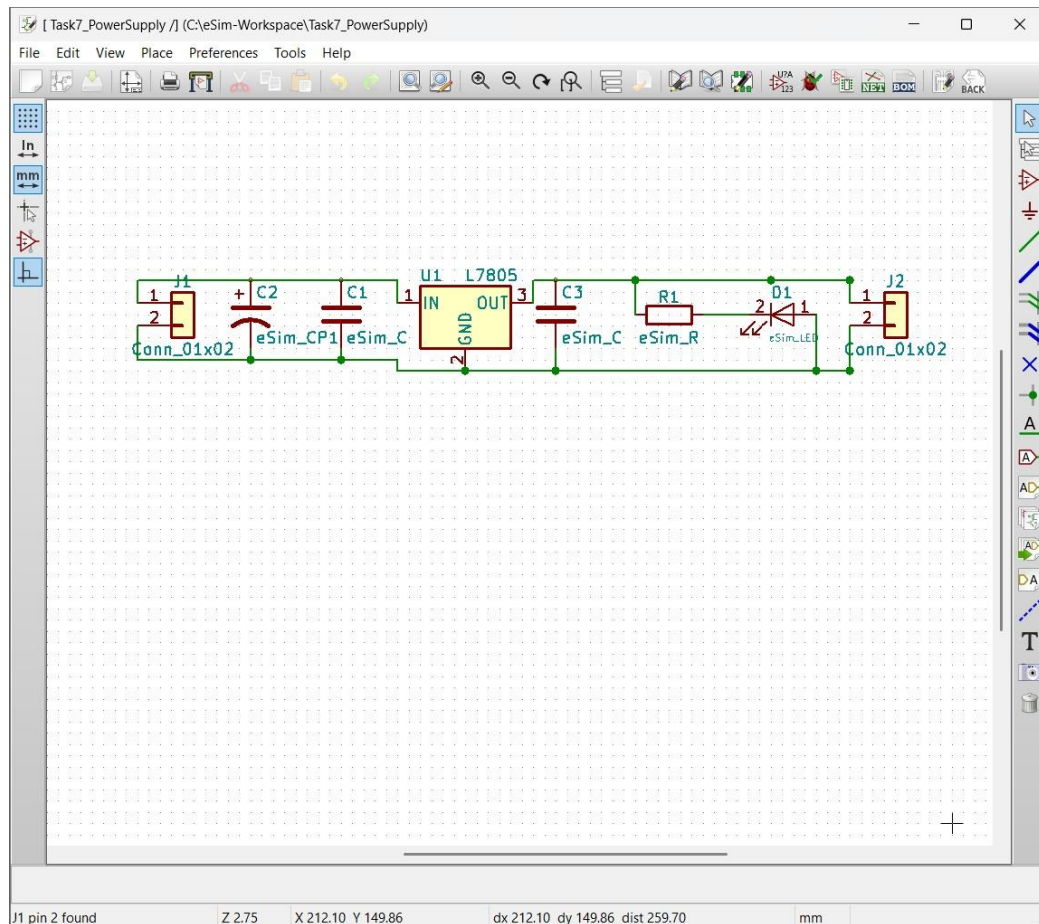


Figure 1: Schematic of the Power Supply Circuit in eSim 2.3

Block Diagram (s): DC Input (J1) → Input Filter Capacitors (C2, C1) → L7805 Voltage Regulator (U1) → Output Filter Capacitor (C3) → LED Indicator Branch (R1 + D1) → Regulated DC Output (J2)

Expected Results (Input, Output waveforms and/or Multimeter readings) : Input: 9V–12V DC (unregulated), with minor ripple depending on the source. Output: a steady 5V DC rail at J2, essentially flat once filtered by C3, capable of supplying up to 1A. A multimeter placed across J2 should read approximately 5.0V DC across the full rated input range and load current, and D1 should light up whenever the output rail is active.

Supporting Evidence: PCB Layout, 3D View & DRC Verification (eSim 2.3)

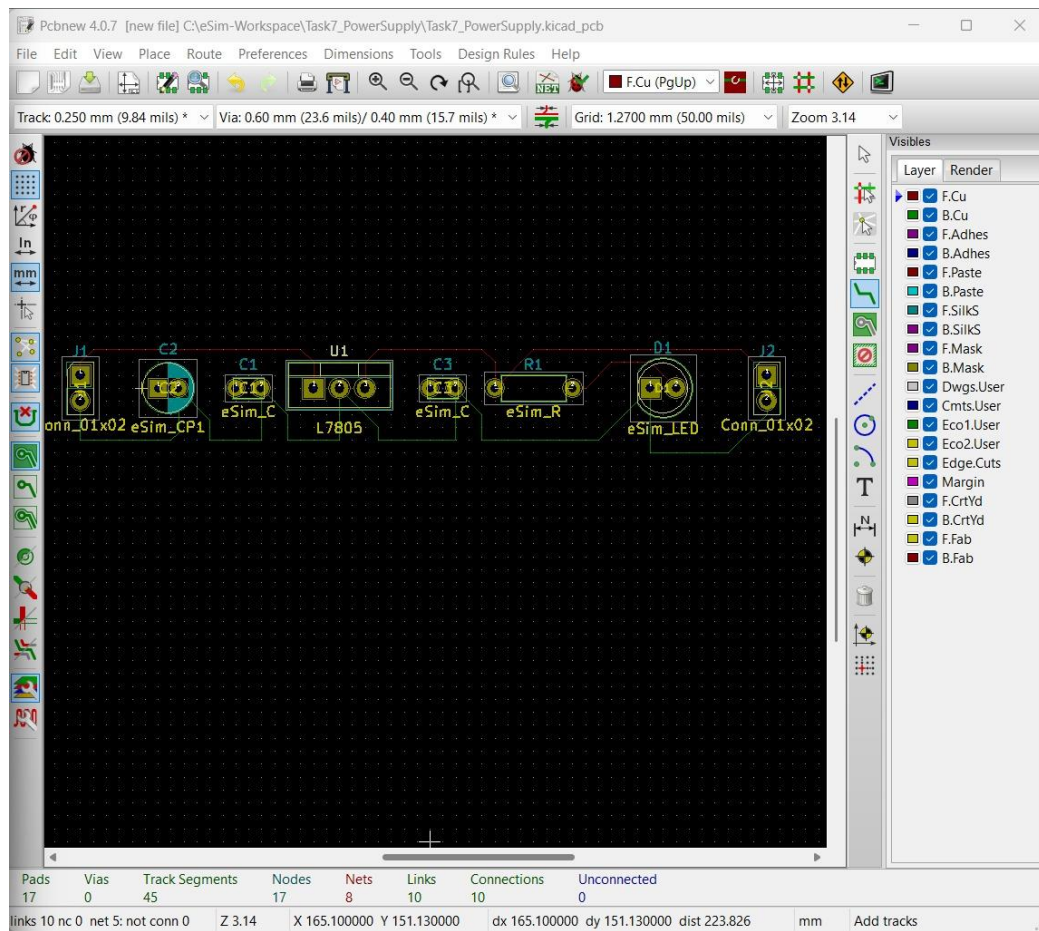


Figure 2: PCB Layout of the Power Supply Circuit

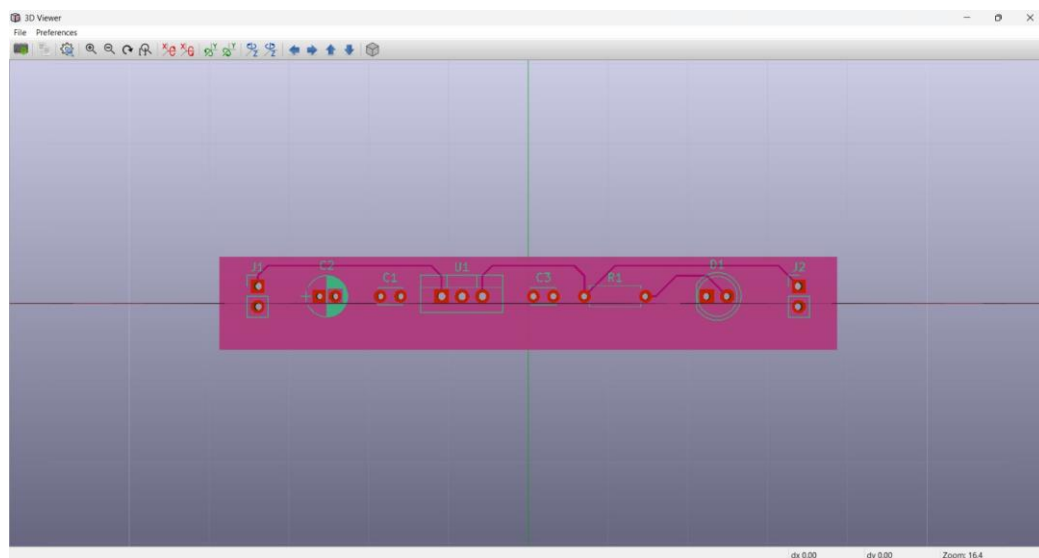


Figure 3: 3D View of the Power Supply PCB

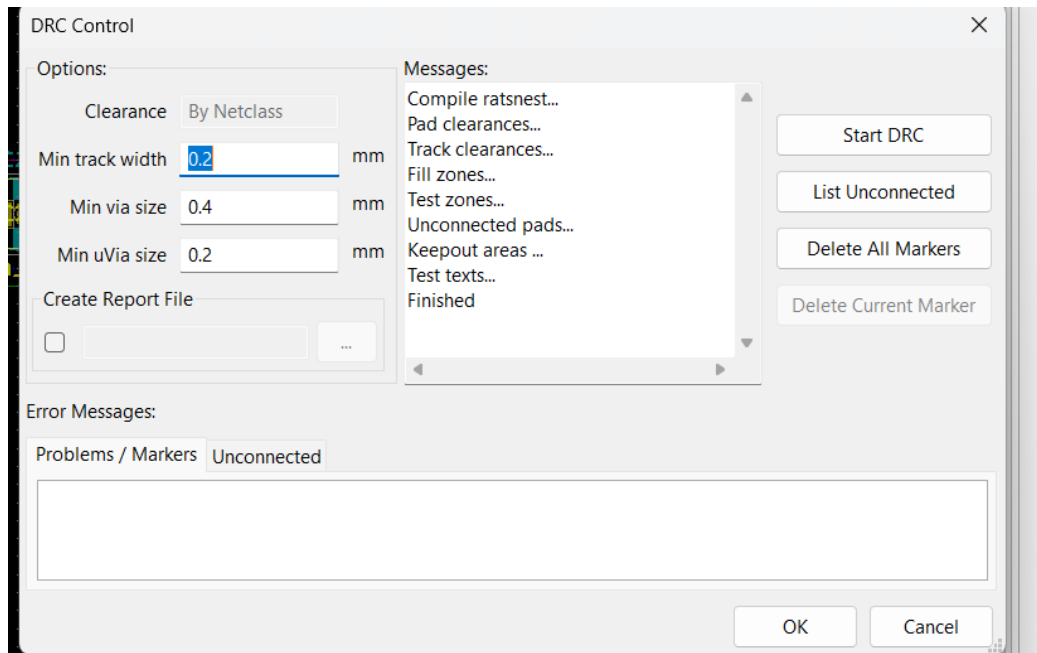


Figure 4: DRC Results -0 Errors

Research Paper/Journal/etc.:

Title: A Low-Voltage, Low Quiescent Current, Low Drop-Out Regulator

Author: Gabriel A. Rincon-Mora, Phillip E. Allen

Page No.: 36–44

Link: <https://ieeexplore.ieee.org/document/654935>

Source/Reference(s): L7805 (78xx series) datasheet, STMicroelectronics/Texas Instruments; eSim 2.3 / KiCad documentation, FOSSEE, IIT Bombay (<https://esim.fossee.in>); course notes on linear voltage regulator design.